A Mini-Project report on

# SOLAR LIGHT DESIGN FOR NIGHT TIME

*Submitted in partial fulfillment of the requirements for the award of the degree of*

## BACHELOR OF TECHNOLOGY
## IN
## ELECTRONICS AND COMMUNICATIONS ENGINEERING

*SUBMITTED BY*


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Aushapur (v), Ghatkesar (m), Medchal Dist, Hyderabad-501301


2021

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# CERTIFICATE

This is to certify that the Mini Project report **"SOLAR LIGHT DESIGN FOR NIGHT TIME"** being submitted by B. Akhil (18P61A0418), Cherupalli Aryan mouli (18P61A0451),CH. Roshini Reddy(18P61A0455), partial fulfillment for the award of the Degree of Bachelor of Technology in **ELECTRONICS & COMMUNICATION ENGINEERING** Jawaharlal Nehru Technological University is a record of a bonafide work carried out by them under my guidance and supervision.

The result embodied in this project report has not been submitted to any other University/Institution for the award of any Degree/Diploma.

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# CANDIDATES DECLARATION

We hereby declare that this Mini Project Report titled **"SOLAR LIGHT DESIGN FOR NIGHT TIME"** submitted by us to the Department of **Electronics and Communication Engineering,** VBIT, Aushapur, Under JNTUH, is a Bonafide work undertaken by and it is not submitted to any other University or Institution for the award of any degree or diploma.

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# ACKNOWLEDGEMENT

At the outset we sincerely thank God for having got my Mini project report completed in time. Firstly, we would thank our parents who have been a motivating factor throughout our lives. Secondly, we sincerely thank our principal **Dr. PVS Srinivas** and our Head of the department **Dr. U. Poorna Lakshmi** for their kind cooperation and Encouragement for the successful completion of Seminar work and providing the necessary facilities.

We are most obliged and grateful to our project guide **Mr. CH. Rajendra Prasad**, for giving us guidance in completing this project successfully.

We express our sincere gratitude to our Project coordinators, Department of ECE and my other faculty for attending our project seminars and for their insightful comments and constructive suggestions to improve the quality of this project work.

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# ABSTRACT

Solar Light Bulb at Night Time is simple yet powerful concept which uses transistor as a switch. By using this system manual works can be eliminated up to 90%. The light bulb can be automatically ON or OFF which happens by setting time. If the battery is fully charged has been constructed. Here, solar energy is collected with the aid of a solar panel and thus, a battery is charged during day time with the help of a simple charging circuit. During night time, this stored energy is used to lights. The device can be used for small-scale lighting applications in remote areas that are far away from the power grid. The system has a panel to collect the sun's energy, a battery to store that energy and a light source to use the energy. There is also a PIR sensor which detects the motion of a person passing under light and useful to auto decrease and increase voltage of light when no one passes and any one passes respectively. The system operates like a bank account. Withdrawals from the battery to power the light source must be compensated for by commensurate deposits of energy form the solar panels. PIR sensor detects a human being moving around within approximately 10m from the sensor. This is an average value, as the actual detection range is between 5m and 12m.PIR are fundamentally made of a pyro electric sensor, which can detect levels of infrared radiation. For numerous essential projects or items that need to discover when an individual has left or entered the area. PIR sensors are incredible, they are flat control and minimal effort, have a wide lens range, and are simple to interface with.

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# CHAPTER-1

## INTRODUCTION

With the intensification of energy crisis all over the world, all the countries are looking for the way to solve this serious problem. One way is to search the new energy and take advantage of the renewable energy. Another way is to exploit the new energy-saving technologies to reduce energy consumption, and improve utilization efficiency of energy. Solar energy is the most direct, common, and clean energy on our planet we have already found until now. Total solar energy absorbed by the Earth is about 3,850,000 exajoules (EJ) in one year, which is even twice as much as all the non-renewable resources on the earth found and used by human being, including coal, oil, natural gas, and uranium etc. The solar resources can be seemed inexhaustible.

LED is a solid state semiconductor device which can convert electrical energy into visible light. It is characterized with small size, low power consumption, long service life, environmental protection and durance. The spectrum of the LED is a mostly concentrated in the visible light spectrum, so it has a high luminous efficiency which can be described as the great reform in the solid light source.

This essay briefly describes the solar led street lighting system. It uses the solar radiation energy to charge the battery with the solar panel during day time, and offer energy to the LED light equipment at night. This system has a double advantage in both utilization of new energy and energy-saving.



## OBJECTIVE:

To create a device that helps in compensating the Growing Energy Demand by using renwable energy source as Solar.

# CHAPTER-2

# EMBEDDED SYSTEM

## 2.1-INTRODUCTION TO EMBEDDED SYSTEMS:

Our day-to-day life is becoming more and more dependent on "Embedded Systems and Digital Technologies. Embedded technologies are bonding into our daily activities even without our knowledge. We know the fact that the refrigerator, washing machine, microwave oven, air conditioner, television, DVD players, and music systems that we use in our home are built around an embedded system. It is nothing but an intelligent embedded system. In your vehicle itself the presence of specialized embedded systems vary from intelligent head lamp controllers, engine controllers and ignition control Systems to complex air bag control systems to protect you from a severe accident.

People experience the power of embedded systems and enjoy the features and comfort provided by them. Most of us are totally unaware or ignorant of the intelligent embedded systems giving us so much comfort and security. Embedded systems are like reliable servants- they don't like to reveal their identity and neither the complaint about their workloads to their owners or bosses. They are always working behind the scenes and are dedicated to their assigned task till their last breath. This book gives you an overview of embedded systems, the various steps involved in their design and development and the major domains where they are deployed.

## 2.2-FEATURES OF EMBEDDED SYSTEMS:

- The versatility of the embedded computer system lends itself to utility in all kinds of enterprises, from the simplification of deliverable products to a reduction in costs in their development and manufacture. Usually a low power consumption CPU with a limited amount of memory is used in embedded systems. Complex systems with rich functionality employ special operating systems that take into account major characteristics of embedded systems. Embedded operating systems have minimized footprint and may follow real-time operating system specifics. The special computers system is usually less powerful than General purpose systems,although some expectations do exist where embedded systems are very powerful and complicated. Usually a low

power consumption CPU with a limited amount of memory is used in embedded systems. Some embedded systems have to operate in extreme environment conditions such as very high temperature & humidity.

- Many embedded systems use very small operating systems; most of these provide very limited operating system capabilities. Since the embedded system is dedicated to specific tasks, design engineers can optimize it, reducing the size and cost of the product, or increasing the reliability and performance some embedded systems are mass-produced, benefiting from economies of scale. Some embedded systems have to operate in extreme environment conditions such as very high temperature & humidity. For high volume systems such as portable music players or mobile phones, minimizing cost is usually the primary design consideration. Engineers typically select hardware that is just "good enough" to implement the necessary functions. For low volume or prototype embedded systems, general purpose computers may be adapted by limiting the programs or by replacing the operating system with a real-time operating system.

## 2.3-History and Future:

Given the definition of embedded systems earlier is this chapter; the first such systems could not possibly have appeared before 1971. That was the year Intel introduced the world's first microprocessor. This chip, the 4004, was designed for use in a line of business calculators produced by the Japanese Company Busicom. In 1969, Busicom asked Intel to design a set of custom integrated circuits-one for each of their new calculator models. The 4004 was Intel's response rather than design custom hardware for each calculator, Intel proposed a general-purpose circuit that could be used throughout the entire line of calculators. Intel's idea was that the software would give each calculator its unique set of features.

The microcontroller was an overnight success, and its use increased steadily over the next decade. Early embedded applications included unmanned space probes, computerized traffic lights, and aircraft flight control systems. In the 1980s, embedded systems quietly rode the waves of the microcomputer age and brought microprocessors into every part of our kitchens (bread machines, food processors, and microwave ovens), living rooms (televisions, stereos, and remote controls), and workplaces (fax machines, pagers, laser printers, cash registers, and credit card readers).

It seems inevitable hat the number of embedded systems will continue to increase rapidly. Already there are promising new embedded devices that have enormous market potential; light switches and thermostats that can be central computer, intelligent air-bag systems that don't inflate when children or small adults are present, pal-sized electronic organizers and personal digital assistants (PDAs), digital cameras, and dashboard navigation systems. Clearly, individuals who possess the skills and desire to design the next generation of embedded systems will be in demandfor quite some time.

## 2.4-CHARACTERISTICS OF EMBEDDED SYSTEMS:

Embedded computing systems generally exhibit rich functionality complex functionality isusually the reason for introducing cups into the design. However, they also exhibit many non- functional requirements that make the task especially challenging:

- Real-time deadlines that will cause system failure if not met.
- Multi-rate operation.
- In many cases, low power consumption.
- Low manufacturing cost, which often means limited code size.

Workstation programmers often concentrate on functionality. They may consider the performance characteristics of a few computational kernels of their software, but rarely analyze the total application. They almost never consider power consumption and manufacturing cost. The need to juggle all these requirements makes embedded system programming very challenging and is the reason why embedded system designers need to understand computer architecture.

## 2.5-TYPES OF EMBEDDED SYSTEM:

Based on functionality and performance embedded systems categorized as 4 types

1. Standalone embedded systems.
2. Real time embedded systems.
3. Networked information appliances.
4. Mobile devices.

1. **Standalone embedded systems:** As the name implies, standalone systems work in standalone mode. They take i/p. Process them and produce the desire o/p. The i/p can be an electrical signal from transducer or temperature signal or commands from human being. The o/p can be electrical signal to drive another system an led or LCD display.
Ex digital camera, microwave oven, CD player, air conditioner etc.

2. **Real time embedded systems**: In this type of an embedded system a specific workhas to be complete in a particular period of time.
Hard real time systems: - embedded real time used in missiles.Soft real time systems: - DVD players.

3. **Networked information appliances:** Embedded systems that are provided with n/w interfaces and accessed by n/w such as local area n/w or internet are called network information appliances.

EX. a web camera is connected to the internet. Camera can send pictures in real time to any computers connected to the internet.

4. **Mobile devices**: Actually it is a combination of both VLSI and Embedded system Mobile devices such as mobile phone, personal digital assistants, smart phones etc. are special category of embedded system. Embedded systems are based on the concept of the microcontroller, a single integrated circuit that contains all the technology required to run an application.

- Central processing Unit.
- Input/output interfaces (such as serial ports).
- Peripherals (such as timers).
- ROM, EEPROM (or) Flash memory for program storage.
- RAM for data storage Clock generator.

By integrating all of these features into a single chip it is possible to greatly reduce the number of chips and wiring necessary to control an electronic device, dramatically reducing complexity, size and cost.

- **Size & Weight**: Microcontrollers are designed to deliver maximum performance for Minimum size and weight. A centralized on-board computer system would greatly outweigh a collection of microcontrollers.
- **Efficiency**: Microcontrollers are designed to perform repeated functions for long periods of time without failing or requiring service.
- **Micro Controller**: It is a chip through which we can connect many other devices and also those are controlled by the program the program which burn into it. Here we are using Arduino as the main device to control.

## 2.6-Real Time Systems:

One subclass of embedded is worthy of an introduction at this point. As commonly defined, a real-time system is a computer system that has timing constraints. In other words, a real-time system is partly specified in terms of its ability to make certain calculations or decisions in a timely manner. These important calculations are said to have deadlines for completion. And, for all practical purposes, a missed deadline is just as bad as a wrong answer.

The issue of what if a deadline is missed is a crucial one. For example, if the real-time system is

part of an airplane's flight control system, it is possible for the lives of the passengers and crew to be endangered by a single missed deadline. However, if instead the system is involved in satellite communication, the damage could be limited to a single corrupt data packet. The more severe the consequences, the more likely it will be said that the deadline is "hard" and thus, the System is a hard real-time system. Real-time systems at the other end of this discussion are said to have "soft" deadlines.

All of the topics and examples presented in this book are applicable to the designers of real-time system who is more delight in his work. He must guarantee reliable operation of the software and hardware under all the possible conditions and to the degree that human lives depend upon three system's proper execution, engineering calculations and descriptive paperwork.

**Application Areas:** Nearly 99 per cent of the processors manufactured end up in embedded systems. The embedded system market is one of the highest growth areas as these systems are used in very market segment- consumer electronics, office automation, industrial automation, biomedical engineering, wireless communication, Data communication, telecommunications, transportation, military and so on.

**Consumer appliances:** At home we use a number of embedded systems which include digital camera, digital diary, DVD player, electronic toys, microwave oven, remote controls for TV and air- conditioner, VCO player, video game consoles, video recorders etc. Today’s high-tech car has about 20 embedded systems for transmission control, engine spark control, air-conditioning, navigation etc. Even wristwatches are now becoming embedded systems. The palmtops are powerful embedded systems using which we can carry out many general-purpose tasks such as playing games and word processing.

**Office automation:** The office automation products using em embedded systems are copying machine, fax machine, key telephone, modem, printer, scanner etc.

**Industrial automation:** Today a lot of industries use embedded systems for process control. These include pharmaceutical, cement, sugar, oil exploration, nuclear energy, electricity generation and transmission. The embedded systems for industrial use are designed to carry out specific tasks such as monitoring the temperature, pressure, humidity, voltage, current etc., and then take appropriate action based on the monitored levels to control other devices or to send information to a centralized monitoring station. In hazardous industrial environment, where

Human presence has to be avoided; robots are used, which are programmed to do specific jobs. The robots are now becoming very powerful and carry out many interesting and complicated tasks such as hardware assembly.

**Medical electronics:** Almost every medical equipment in the hospital is an embedded system. Theseequipment's include diagnostic aids such as ECG, EEG, blood pressure measuring devices, X-ray scanners; equipment used in blood analysis, radiation, colonoscopy, endoscopy etc. Developments in medical electronics have paved way for more accurate diagnosis of diseases.

**Computer networking:** Computer networking products such as bridges, routers, Integrated ServicesDigital Networks (ISDN), Asynchronous Transfer Mode (ATM), X.25 and frame relay switches are embedded systems which implement the necessary data communication protocols. For example, a router interconnects two networks. The two networks may be running different protocol stacks. The router's function is to obtain the data packets from incoming pores, analyze the packets and send them towards the destination after doing necessary protocol conversion. Most networking equipments, other than the end systems (desktop computers) we use to access the networks, are embedded systems.

**Telecommunications:** In the field of telecommunications, the embedded systems can be categorizedas subscriber terminals and network equipment. The subscriber terminals such as key telephones, ISDN phones, terminal adapters, web cameras are embedded systems. The network equipment includes multiplexers, multiple access systems, Packet Assemblers Dissemblers (PADs), sate11ite modems etc. IP phone, IP gateway, IP gatekeeper etc. are the latest embedded systems that provide very low-cost voice communication over the Internet.

**Wireless technologies:** Advances in mobile communications are paving way for many interesting applications using embedded systems. The mobile phone is one of the marvels of the last decade of the 20'h century. It is a very powerful embedded system that provides voice communication while we are on the move. The Personal Digital Assistants and the palmtops can now be used to access multimedia services over the Internet. Mobile communication infrastructure such as base station controllers, mobile switching centers are also powerful embedded systems.

**Insemination:** Testing and measurement are the fundamental requirements in all scientific and engineering activities. The measuring equipment we use in laboratories to measure parameters such as weight, temperature, pressure, humidity, voltage, current etc. are all embedded systems. Test equipment such as oscilloscope, spectrum analyzer, logic analyzer, protocol analyzer, radio communication test set etc. are embedded systems built around powerful processors. Thank to miniaturization, the test and measuring equipment are now becoming portable facilitating easy testing and measurement in the field by field-personnel.

**Security:** Security of persons and information has always been a major issue. We need to protect ourhomes and offices; and also the information we transmit and store. Developing embedded systemsfor security applications is one of the most lucrative businesses nowadays. Security devices at homes,offices, airports etc. for authentication and verification are embedded systems. Encryption devices arenearly 99 per cent of the processors that are manufactured end up in embedded systems. Embedded systems find applications in. Every industrial segment-consumer electronics, transportation, avionics,biomedical engineering, manufacturing, process control and industrial automation, data communication, telecommunication, defense, security etc. Used to encrypt the data/voice being transmitted on communication links such as telephone lines. Biometric systems using fingerprint and face recognition are now being extensively used for user authentication in banking applications as well as for access control in high security buildings.

**Finance:** Financial dealing through cash and cheques are now slowly paving way for transactions using smart cards and ATM (Automatic Teller Machine, also expanded as Any Time Money) machines. Smart card, of the size of a credit card, has a small micro-controller and memory; and it interacts with the smart card reader! ATM machine and acts as an electronic wallet. Smart card technology has the capability of ushering in a cashless society. Well, the list goes on. It is no exaggeration to say that eyes wherever you go, you can see, or at least feel, the work of an embedded system.

## 2.7-Overview of Embedded System Architecture:

Every embedded system consists of custom-built hardware built around a Central Processing Unit (CPU). This hardware also contains memory chips onto which the software is loaded. The software residing on the memory chip is also called the 'firmware'. The embedded system architecture can be represented as a layered architecture as shown in Fig.

1

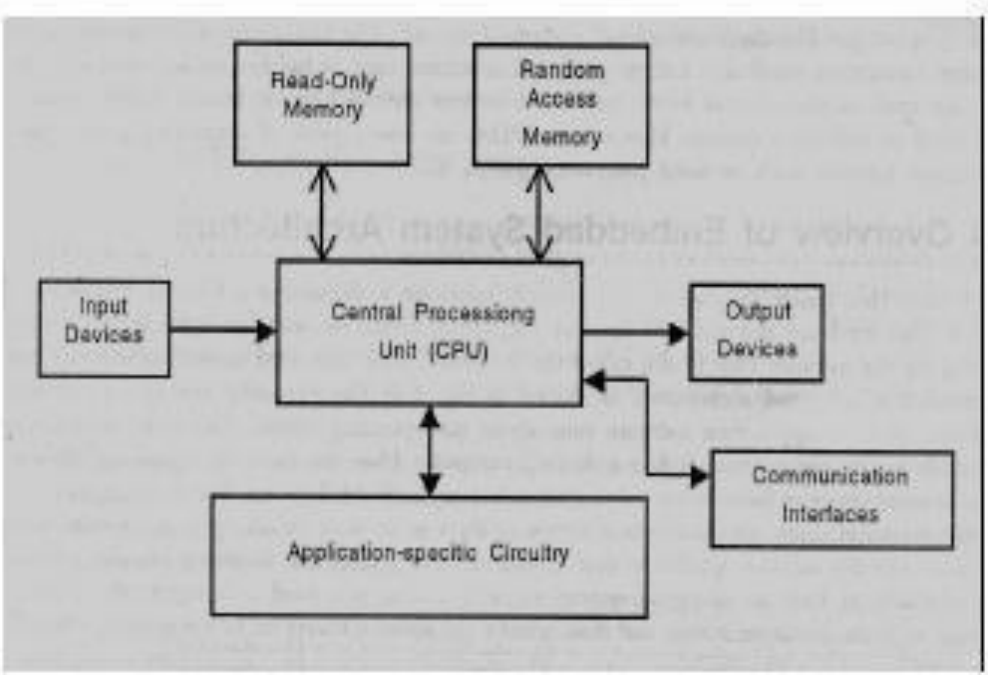



**Fig** 2.1 Blocks of hardware embedded system

## Central Processing Unit (CPU):

The Central Processing Unit (processor, in short) can be any of the following: microcontroller, microprocessor or Digital Signal Processor (DSP). A micro-controller is a low-cost processor. Its main attraction is that on the chip itself, there will be many other components such as memory, serial communication interface, analog-to digital converter etc. So, for small applications, a micro-controller is the best choice as the number of external components required will be very less. On the other hand, microprocessors are more powerful, but you need to use many external components with them. D5P is used mainly for applications in which signal processing is involved such as audio and video processing.

**Memory:**

The memory is categorized as Random Access 11emory (RAM) and Read Only Memory (ROM).The contents of the RAM will be erased if power is switched off to the chip, whereas ROM

retains the contents even if the power is switched off. So, the firmware is stored in the ROM. When power isswitched on, the processor reads the ROM; the program is program is

Executed.

**Input devices:**

Unlike the desktops, the input devices to an embedded system have very limited capability. There will be no keyboard or a mouse, and hence interacting with the embedded system is no easy task. Many embedded systems will have a small keypad-you press one key to give a specific command.A keypad may be used to input only the digits. Many embedded systems used in process controldo not have any input device for user interaction; they take inputs from sensors or transducers 1'fnd produce electrical signals that are in turn fed to other systems.

**Output devices:**

The output devices of the embedded systems also have very limited capability. Some embedded systems will have a few Light Emitting Diodes (LEDs) to indicate the health status of the system modules, or for visual indication of alarms. A small Liquid Crystal Display (LCD) may also be used to display some important parameters.

**Communication interfaces:**

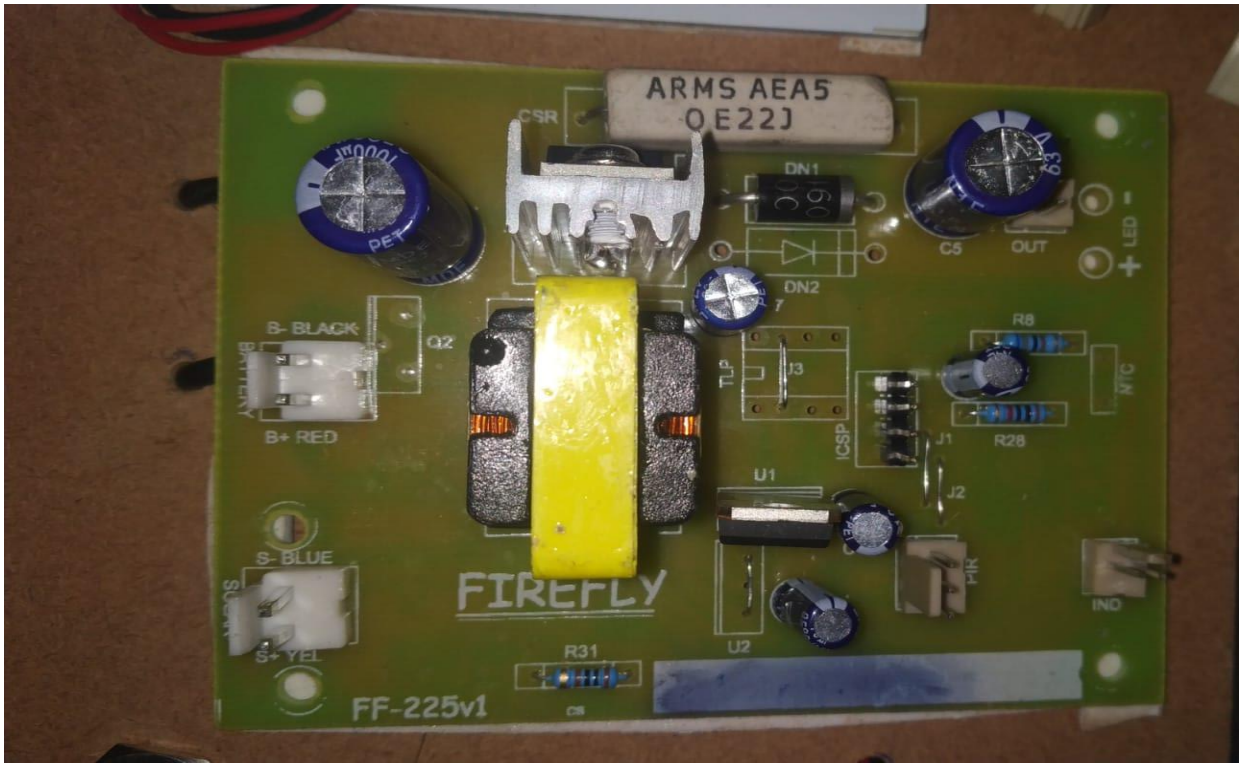
The embedded systems may need to, interact with other embedded systems at they may have to transmit data to a desktop. To facilitate this, the embedded systems are provided with one or a few communication interfaces such as RS232, RS422, RS485, Universal Serial Bus (USB), and IEEE 1394, Ethernet etc.

# CHAPTER-3

## ANALYSIS

## HARDWARE REQUIREMENTS

### CHARGE CONTROLLER




Solar charge controllers regulate the energy flowing from the PV array and transfer it directly to the batteries as a DCcoupled system, which is the most efficient and effective manner. Controller is a very significant device in solar streetlight, used to decide the status of the charging and lighting by switch on or switch off. Some recent controllers are preprogrammed and it consists of a battery charger, a Led lamp driver, a driver, a secondary power supply, an MCU and a protection circuit. The battery can be controlled by the controller from the under and over charging conditions. The battery can be charged by the power received from the solar panels in the sunrise and while in the sunset it charges the battery. Some modern controller are programmable so that user can decide the appropriate chance of charging, lighting and dimming.Regulators for solar systems are designed to keep the batteried charged at peak without overcharging.The control of battery charging is so important the most manufactures of high quality batteries specify the requirements

for voltage regulation, low voltage disconnect and temperature compensation. When these limits are not respected, it is common for batteries to fail after less than one quarter of their normal life expectancy, regardless of their quality or their cost. A good charge controller is not expensive in relation to the total cost of a power system.

## PIR SENSOR

A passive infrared sensor is an electronic sensor that measures infrared light radiating from objects in its field of view. They are most often used in PIR-based motion detectors.

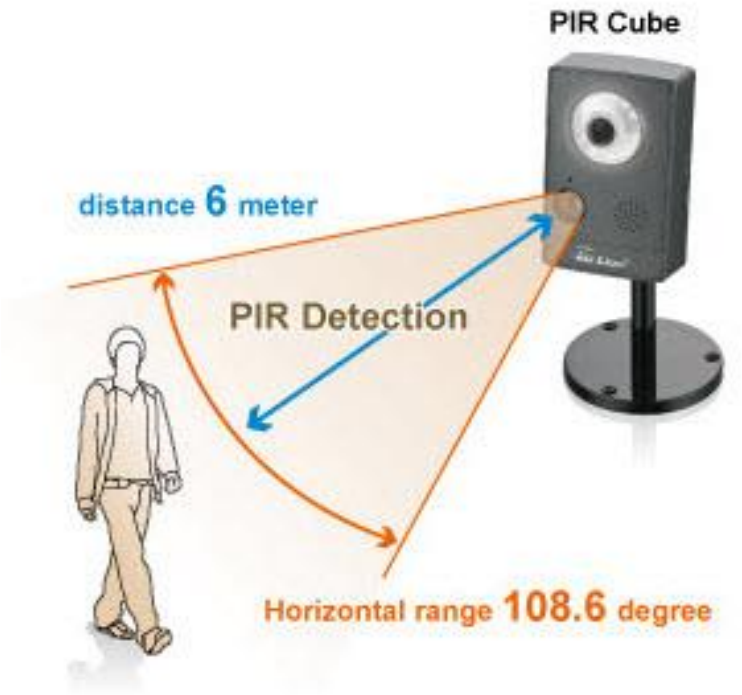
PIR sensors are commonly used in security alarms and automatic lighting applications.



Technically, PIR is made of a pyroelectric sensor, which is able to detect different levels of infrared radiation. For example, Everything emits varied level radiation and the level of radiation will increase with the increase of the object's temperature.

PIR sensors are also known as PID or Passive Infrared Detectors. Thus, the PIR sensor can detect infrared radiation that is emitted by particles.Generally, PIR can detect animal/human movement in a requirement range, which is determined by the spec of the specific sensor. The detector itself does not emit any energy but passively receives it, detects infrared radiation from the environment.

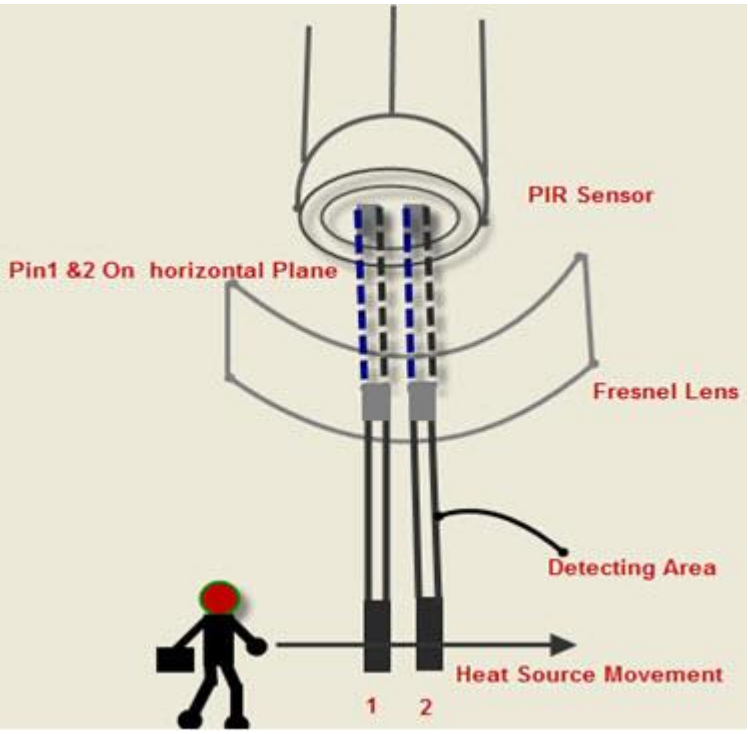
.




The PIR sensor is pretty complicated when compared to the other sensors. Since they have 2 slots, and the slots are made of sensitive material.

The Fresnel lens is used to see that the two slots of the PIR can see out past some distance. When the sensor is inactive, then the two slots senses the same amount of IR. The ambient amount radiates from the outdoors, walls or room, etc.

When a human body or any animal passes by, it intercepts the first slot of the PIR sensor. This causes a positive differential change between the two bisects. But when the body leaves the sensing area, the sensor generates a negative differential change between the two bisects.




**Range of Different PIR Sensors**

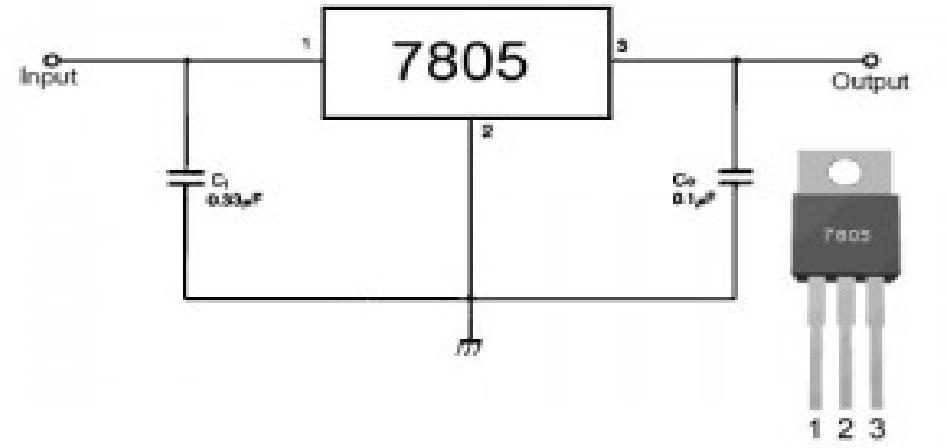
Indoor passive infrared: Detection distances range from 25 cm to 20 m.

Indoor curtain-type: The detection distance ranges from 25 cm to 20 m.

Outdoor passive infrared: The detection distance ranges from 10 meters to 150 meters.

Outdoor passive infrared curtain detector: distance from 10 meters to 150 meters

# VOLTAGE REGULATOR 7805




**Features**

- Output Current up to 1A
- Output Voltages of 5, 6, 8, 9, 10, 12, 15, 18, 24V
- Thermal Overload Protection
- Short Circuit Protection
- Output Transistor Safe Operating Area Protection

## Description

TheLM78XX/LM78XXA series of three-terminal positive regulators are available in the TO-220/D-PAK package and with several fixed output voltages, making them useful in a Wide range of applications. Each type employs internal current limiting, thermal shutdown and safe operating area protection, making it essentially indestructible. If adequate heat sinking is provided, they can deliver over 1A output Current. Although designed primarily as fixed voltage regulators, these devices can be used with external components to obtain adjustable voltages and currents.

A voltage regulator is an electrical regulator designed to automatically maintain a constant voltage level. A voltage regulator may be a simple "feed-forward" design or may include feedback control. It may use an electromechanical mechanism, or electronic components. Depending on the design, it may be used to regulate one or more AC or DC voltages.

Electronic voltage regulators are found in devices such as computer power supplies where they stabilize the DC voltages used by the processor and other elements. In automobile alternators and central power station generator plants, voltage regulators control the output of the plant. In an electric power distribution system, voltage regulators may be installed at a substation or along distribution lines so that all customers receive steady voltage independent of how much power is drawn from the line. The output voltage can only be held roughly constant; the regulation is specified by two measurements:Load regulation is the change in output voltage for a given change in load current (for example: "typically 15mV, maximum 100mV for load currents between 5mA and 1.4A, at some specified temperature and input voltage").Line regulation or input regulation is the degree to which output voltage changes with input (supply) voltage changes - as a ratio of output to input change (for example "typically 13mV/V"), or the output voltage change over the entire specified input voltage range (for example "plus or minus 2% for input voltages between 90V and 260V, 50-60Hz").

Other important parameters are:

- Temperature coefficient of the output voltage is the change in output voltage with temperature (perhaps averaged over a given temperature range), while...
- Initial accuracy of a voltage regulator (or simply "the voltage accuracy") reflects the error in output voltage for a fixed regulator without taking into account temperature or aging effects on output accuracy.
- Dropout voltage is the minimum difference between input voltage and output voltage for which the regulator can still supply the specified current. A Low Drop-Out (LDO) regulator is designed to work well even with an input supply only a Volt or so above the output voltage.
- Absolute maximum ratings are defined for regulator components, specifying the continuous and peak output currents that may be used (sometimes internally limited), the maximum input voltage, maximum power dissipation at a given temperature, etc.

Electronic voltage regulators are found in devices such as computer power supplies where they stabilize the DC voltages used by the processor and other elements. In automobile alternators and central power station generator plants, voltage regulators control the output of the plant. In an electric power distribution system, voltage regulators may be installed at a substation or along distribution lines so that all customers receive steady voltage independent of how much power is drawn from the line. The output voltage can only be held roughly constant; the

regulation is specified by two measurements:

- Load regulation is the change in output voltage for a given change in load current (for example: "typically 15mV, maximum 100mV for load currents between 5mA and 1.4A, at some specified temperature and input voltage").
- Line regulation or input regulation is the degree to which output voltage changes with input (supply) voltage changes - as a ratio of output to input change (for example "typically 13mV/V"), or the output voltage change over the entire specified input voltage range (for example "plus or minus 2% for input voltages between 90V and 260V, 50-60Hz").

Other important parameters are:

- Temperature coefficient of the output voltage is the change in output voltage with temperature (perhaps averaged over a given temperature range), while...
- Initial accuracy of a voltage regulator (or simply "the voltage accuracy") reflects the error in output voltage for a fixed regulator without taking into account temperature or aging effects on output accuracy.
- Dropout voltage is the minimum difference between input voltage and output voltage for which the regulator can still supply the specified current. A Low Drop-Out§ (LDO) regulator is designed to work well even with an input supply only a Volt§ or so above the output voltage.
- Absolute maximum ratings are defined for regulator components, specifying the continuous and peak output currents that may be used (sometimes internally limited), the maximum input voltage, maximum power dissipation at a given temperature, etc.

- Output noise (thermal white noise§) and output dynamic impedance may be specified as graphs versus frequency, while output ripple noise (mains "hum" or switch-mode "hash" noise) may be given as peak-to-peak or RMS§ voltages, or in terms of their spectra.
- Quiescent current in a regulator circuit is the current drawn internally, not available to the load, normally measured as the input current while no load is connected (and hence a source of inefficiency; some linear regulators are, surprisingly, more efficient at very low current loads than switch-mode designs because of this).
- Transient response is the reaction of a regulator when a (sudden) change of the load current (called the load transient) or input voltage (called the line transient) occurs. Some regulators will tend to oscillate or have a slow response time which in some cases might lead to undesired results. This value is different from the regulation parameters, as that is the stable situation definition. The transient response shows the behaviour of the regulator on a change. This data is usually provided in the technical documentation of a regulator and is also dependent on output capacitan

## LED(LIGHT EMITTING DIODE)



LEDs are semiconductor devices. Like transistors, and other diodes, LEDs are made out of silicon. What makes an LED give off light are the small amounts of chemical impurities that are added to the silicon, such as gallium, arsenide, indium, and nitride. When current passes through the LED, it emits photons as a byproduct. Normal light bulbs produce light by heating a metal filament until its white hot. Because LEDs produce photons directly and not via heat, they are far more efficient than incandescent bulbs. Not long ago LEDs were only bright enough to be used as indicators on dashboards or electronic equipment. But recent advances have made LEDs bright enough to rival traditional lighting technologies. Modern LEDs can replace incandescent bulbs in almost any application.

LEDs are based on the semiconductor diode. When the diode is forward biased (switched on), electrons are able to recombine with holes and energy is released in the form of light. This effect is called electroluminescence and the color of the light is determined by the energy gap of the semiconductor. The LED is usually small in area (less than 1 mm2) with integrated optical components to shape its radiation pattern and assist in reflection.

LEDs present many advantages over traditional light sources including lower energy consumption, longer lifetime, improved robustness, smaller size and faster switching. However, they are relatively expensive and require more precise current and heat management than traditional light sources.

Applications of LEDs are diverse. They are used as low-energy and also for replacements for traditional light sources in well-established applications such as indicators and automotive lighting. The compact size of LEDs has allowed new text and video displays and sensors to be developed, while their high switching rates are useful in communications technology. So here the role of LED is to indicate the status of the components like relays and power circuits.

# SOLAR PANELS

Solar photovoltaic systems use solar energy to produce electricity. The term photovoltaic is composed of "photo", the Greek root for "light", and "volt", a common measurement of electricity named after Alessandro Volta, a scientist renowned for his research on electricity. Together, these terms literally mean "light electricity". Photovoltaic technology can be referred to in short as photovoltaics or PV.hit by Photovoltaic technology relies on the electrical properties of certain materials known as semiconductors. When sunlight, a semiconductor material responds by creating an electrical charge which can then be transferred to anything that uses electricity. These semiconductors are produced in the form of cells, which can then be assembled in groups in a panel. There are many different types of panels available, and each has its particular advantages**.**



Individual panels are often used to charge batteries that power small or remote electric equipment. Depending on the amount of electricity needed, these panels can then be connected in an array to provide larger amounts of electricity to a building or other large user of electricity.

Photovoltaic cells and panels can be manufactured and installed at almost any scale, and as a result are used to power a broad variety of applications. At its smallest, photovoltaic technology powers calculators, laptop computers and other appliances that run on batteries. At its largest, it powers homes, offices and other buildings that use large amounts of electricity, and can be connected to utilities to increase the diversity of our collective electricity supply.

In connecting a photovoltaic system to an end use, several additional structures and technologies are needed. While photovoltaics can be mounted on roofs, it is important to consider the angle at which they face the sun. To transfer electricity to its end use, photovoltaics are connected through intermediary technologies that condition and modify the electricity they produce. These considerations are known as balance of system components, as they maximize the system's efficiency and allow higher amounts of electricity to reach its end use.

Photovoltaic cells work by transforming the photon energy in solar radiation directly into electrical energy without an intermediate mechanical or thermal process. A photovoltaic cell consists of layers of semiconductor materials in contact with each other and fitted with metallic contacts to transfer the released electrons to the external load. Most commercial photovoltaic cells now available are manufactured from crystalline silicon, which is doped to provide the required semiconductor qualities. This is then fitted with the metallic contacts and encapsulated for protection. PV cells operate on the principle that electricity will flow between two different semiconductors when they are put in contact with each other and exposed to light. By linking a number of these cells together a flow of electricity can be achieved.

Solar panels need to have a lot of surface area that can be pointed towards the Sun as the spacecraft moves. More exposed surface area means more electricity can be converted from light energy from the Sun. Sometimes, satellite scientists purposefully orient the solar panels to "off point," or out of direct alignment from the Sun. This happens if the batteries are completely charged and the amount of electricity needed is lower than the amount of electricity made. The extra power will just be vented by a shunt into space as heat.

# RESISTORS

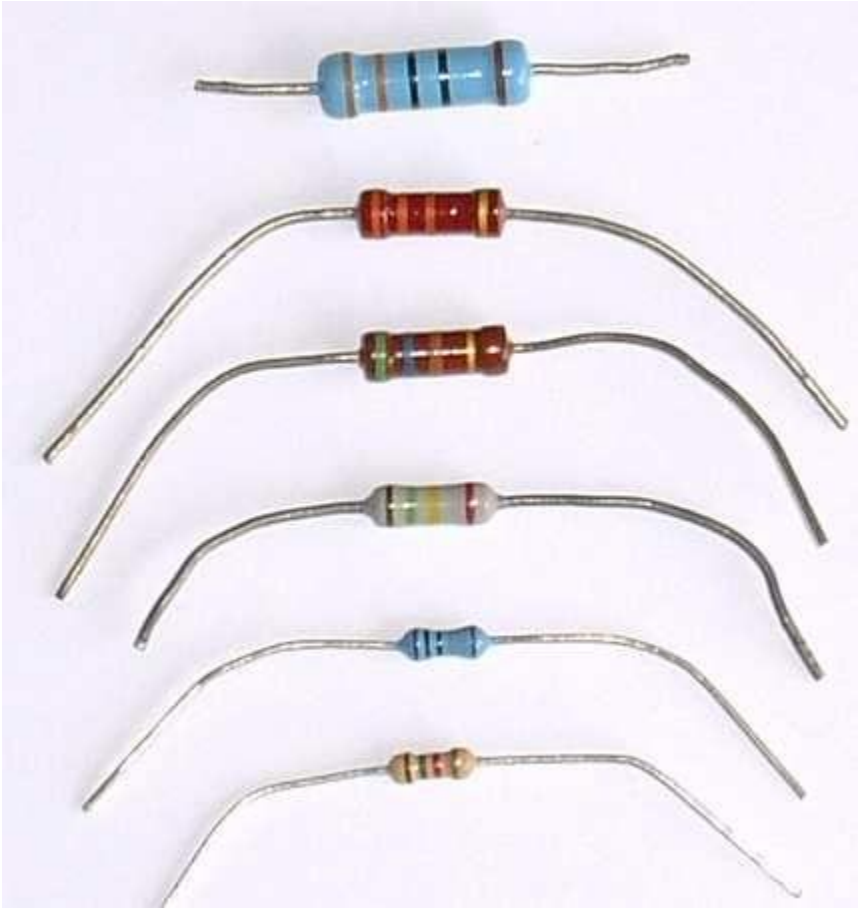
A resistor is a two-terminal electronic component designed to oppose an electric current by producing a voltage drop between its terminals in proportion to the current, that is, in accordance with Ohm's law:

$V = IR$ .

Resistors are used as part of electrical networks and electronic circuits. They are extremely

commonplace in most electronic equipment. Practical resistors can be made of various compounds and films, as well as resistance wire (wire made of a high-resistivity alloy, such as nickel/chrome). The primary characteristics of resistors are their resistance and the power they can dissipate. Other characteristics include temperature coefficient, noise, and inductance. Less well-known is critical resistance, the value below which power dissipation limits the maximum permitted current flow, and above which the limit is applied voltage. Critical resistance depends upon the materials constituting the resistor as well as its physical dimensions; it's determined by design.

Resistors can be integrated into hybrid and printed circuits, as well as integrated circuits. Size, and position of leads (or terminals) are relevant to equipment designers; resistors must be physically large enough not to overheat when dissipating their power.



A resistor is a two-terminal passive electronic component which implements electrical resistance as a circuit element. When a voltage V is applied across the terminals of a resistor, a current I will flow through the resistor in direct proportion to that voltage. The reciprocal of the constant of proportionality is known as the resistance R, since, with a given voltage V, a larger value of R further "resists" the flow of current I as given by Ohm's law.

Resistors are common elements of electrical networks and electronic circuits and are ubiquitous in most electronic equipment. Practical resistors can be made of various compounds and films, as well as resistance wire (wire made of a high-resistivity alloy, such as nickel-chrome). Resistors are also implemented within integrated circuits, particularly analog devices, and can also be integrated into hybrid and printed circuits.

The electrical functionality of a resistor is specified by its resistance: common commercial resistors are manufactured over a range of more than 9 orders of magnitude. When specifying that resistance in an electronic design, the required precision of the resistance may require attention to the manufacturing tolerance of the chosen

resistor, according to its specific application. The temperature coefficient of the resistance may also be of concern in some precision applications. Practical resistors are also specified as having a maximum power rating which must exceed the anticipated power dissipation of that resistor in a particular circuit: this is mainly of concern in power electronics applications. Resistors with higher power ratings are physically larger and may require heat sinking. In a high voltage circuit, attention must sometimes be paid to the rated maximum working voltage of the resistor.

The series inductance of a practical resistor causes its behavior to depart from ohms law; this specification can be important in some high-frequency applications for smaller values of resistance. In a low-noise amplifier or pre-amp the noise characteristics of a resistor may be an issue. The unwanted inductance, excess noise, and temperature coefficient are mainly dependent on the technology used in manufacturing the resistor. They are not normally specified individually for a particular family of resistors manufactured using a particular technology. A family of discrete resistors is also characterized according to its form factor, that is, the size of the device and position of its leads (or terminals) which is relevant in the practical manufacturing of circuits using them.

The ohm (symbol: Ω) is the SI unit of electrical resistance, named after Georg Simon Ohm. An

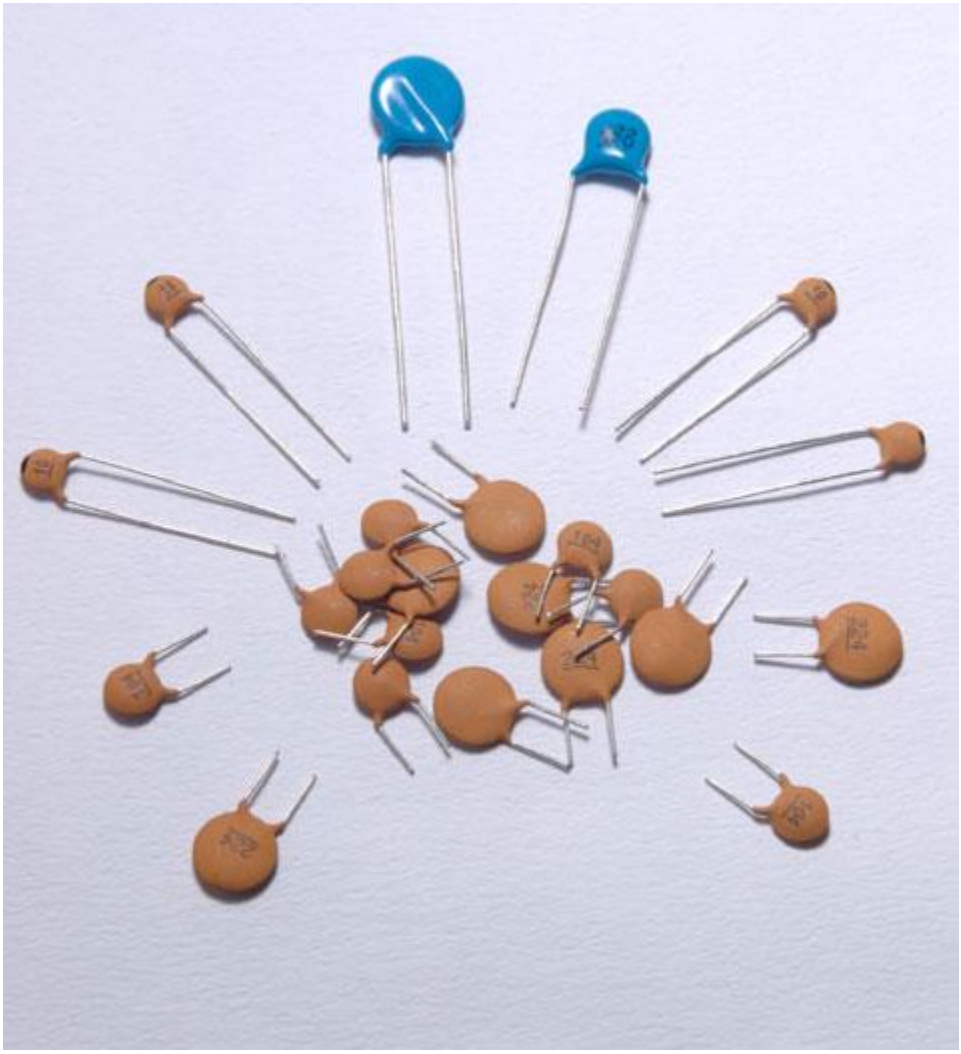
ohm is equivalent to a volt per ampere. Since resistors are specified and manufactured over a very large range of values, the derived units of milliohm (1 mΩ = $10^{-3}$ Ω), kilohm (1 kΩ = $10^{3}$ Ω), and megohm (1 MΩ = $10^{6}$ Ω) are also in common usage.

The reciprocal of resistance R is called conductance G = 1/R and is measured in Siemens (SI unit), sometimes referred to as a mho. Thus a Siemens is the reciprocal of an ohm: $S = \Omega^{-1}$. Although the concept of conductance is often used in circuit analysis, practical resistors are always specified in terms of their resistance (ohms) rather than conductance.

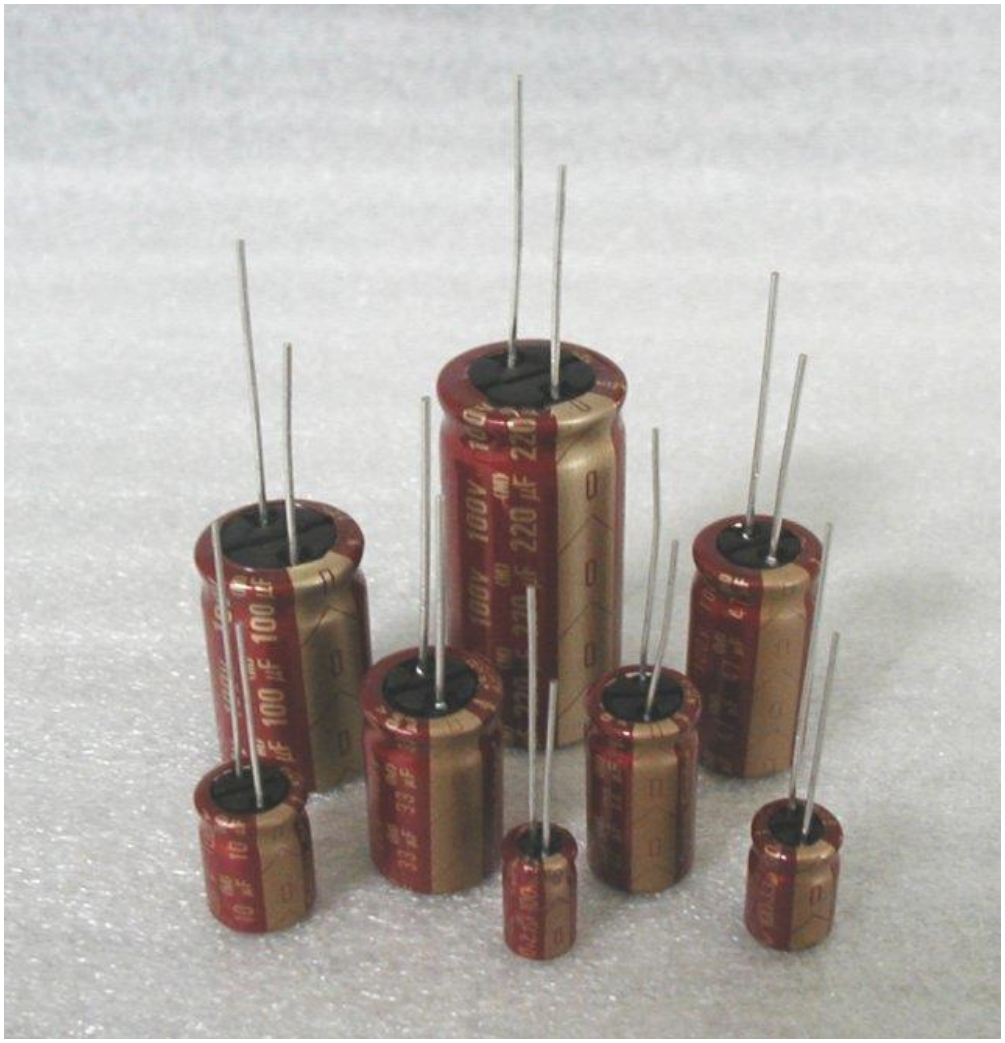
## CAPACITORS

A capacitor or condenser is a passive electronic component consisting of a pair of conductors separated by a dielectric. When a voltage potential difference exists between the conductors, an electric field is present in the dielectric. This field stores energy and produces a mechanical force between the plates. The effect is greatest between wide, flat, parallel, narrowly separated conductors.

An ideal capacitor is characterized by a single constant value, capacitance, which is measured in farads. This is the ratio of the electric charge on each conductor to the potential difference between them. In practice, the dielectric between the plates passes a small amount of leakage current. The conductors and leads introduce an equivalent series resistance and the dielectric has an electric field strength limit resulting in a breakdown voltage.



The properties of capacitors in a circuit may determine the resonant frequency and quality factor of a resonant circuit, power dissipation and operating frequency in a digital logic circuit, energy capacity in a high-power system, and many other important aspects.



A capacitor (formerly known as condenser) is a device for storing electric charge. The forms of practical capacitors vary widely, but all contain at least two conductors separated by a non-conductor. Capacitors used as parts of electrical systems, for example, consist of metal foils separated by a layer of insulating film.

Capacitors are widely used in electronic circuits for blocking direct current while allowing alternating current to pass, in filter networks, for smoothing the output of power supplies, in the resonant circuits that tune radios to particular frequencies and for many other purposes.

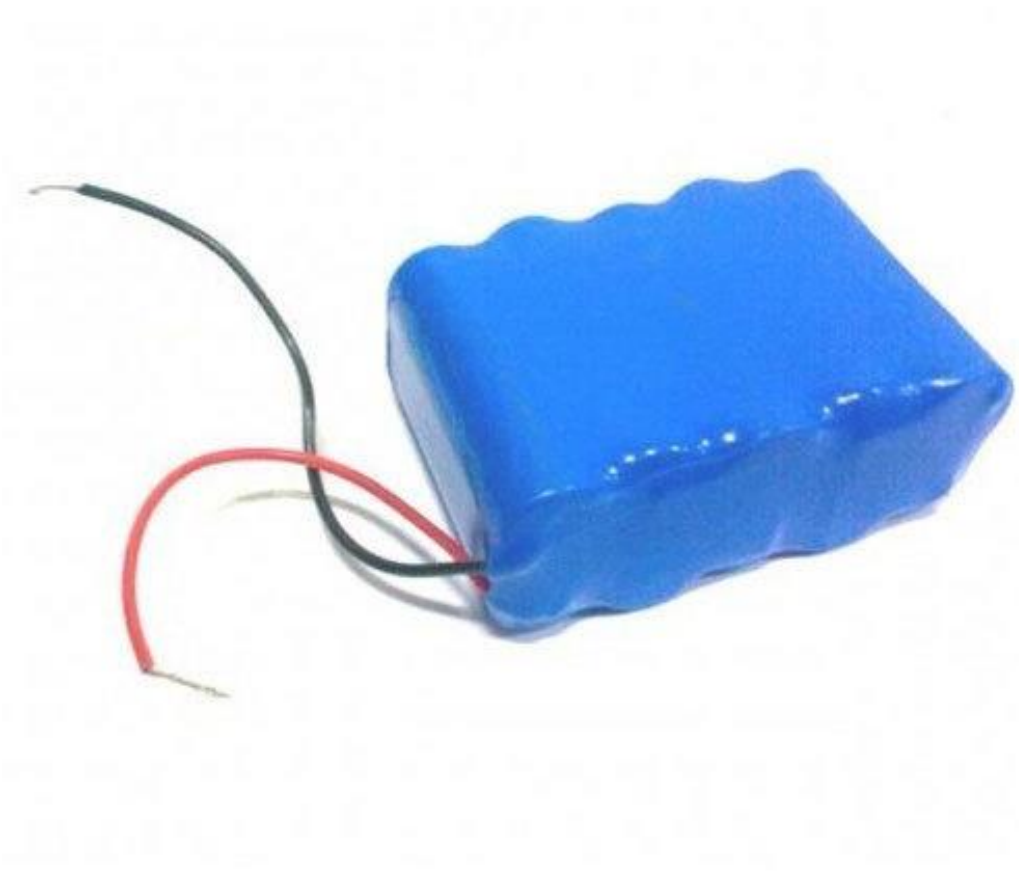
A capacitor is a passive electronic component consisting of a pair of conductors separated by a

dielectric (insulator). When there is a potential difference (voltage) across the conductors, a static electric field develops in the dielectric that stores energy and produces a mechanical force between the conductors. An ideal capacitor is characterized by a single constant value, capacitance, measured in farads. This is the ratio of the electric charge on each conductor to the potential difference between them.

The capacitance is greatest when there is a narrow separation between large areas of conductor, hence capacitor conductors are often called "plates", referring to an early means of construction. In practice the dielectric between the plates passes a small amount of leakage current and also has an electric field strength limit, resulting in a breakdown voltage, while the conductors and leads introduce an undesired inductance and resistance.

## RECHARGEABLE BATTERY

A rechargeable battery, storage, secondary battery or accumulator is a type of electrical battery which can be charged, discharged into a load, and recharged many times, while a non-rechargeable or primary battery is supplied fully charged, and discarded once discharged. Several different combinations of electrode materials and electrolytes are used, including lead-acid, nickel cadmium (NiCad), nickel metal hydride (Ni-MH), lithium ion (Li-ion), and lithium ion polymer (Li-ion polymer). During charging, the battery functions as an electrolytic cell. When the cell is connected to an outside energy source, the electrons in the cathode are forced to flow back to the anode.



**Selection of Battery:**

The requirements of the battery on the solar street light is: slower discharge rate and the long discharge time, it was decided that we usually chose the large-capacity lead-acid batteries, thus the high-energy and maintenance free valve-regulated lead-acid batteries (VRLA) is a better choice. The VRLA battery has a one-way exhaust valve (also called a valve-regulated cap) on the battery cover which can vent surplus gas in case of unusual circumstances. The so called "maintenance-free " is regularly compared with maintenance of the traditional lead-acid battery. The VRLA batteries do not require adding water or acid during service life and it is not necessary to check the electrolyte level. In addition, the standard to measure the quality of battery is reflected as the following aspects:

i Long life as long as life of the system.

ii High reserve capacity and high capacity retention rate, in order to meet the lighting needs in the consecutive rainy period.

iii High charge acceptance efficiency which can maximize output power of the solar cell, and also shorten the charging time.

iv Good sealing performance, no acid or gas leakage, so the battery can be set with controller.

v Wide working temperature range (- 20℃ to 50℃)

vi Good discharge performance The quality of the battery is directly related to performance of the streetlights. Now there is a lot of supporting battery products on the market which are designed for the solar lighting system, we should find the appropriate battery through comparing and testing.

**Capacity calculation of battery:**

The batteries are the main components in the solar LED street lights system, they can store energy which are generated by the solar cell during day time, and meet the power consumption of lighting at night and lighting needs in consecutive rainy days. It is not possible to meet the needs of night lighting if the battery capacity is too small. Inversely If the battery capacity is too large, we need a large solar panels to ensure the battery is fully charged in a limited time during the day . The over-large panels and battery will cause increasing of cost and also the waste. If the solar panel is not large enough, the battery cannot be fully charged in limited period of time during the day . The over-large panels and battery will cause increasing of cost and also the waste. If the solar panel is not large enough, the battery can not be fully charged in limited period of time during the day, it will always be in a state of power deficit, this is a bad effect of the battery life.

**TECHNICAL SPECIFICATIONS OF BATTERY:**

| S.NO | DESCRIPTION | REQUIREMENT |
|---|---|---|
| 1. | Rechargeable Battery | Yes |
| 2. | Type | LiFepo4 |
| 3. | Battery Capacity | 18AH |
| 4. | Battery Voltage | 12.8V |
| 5. | Wattage | 230Wh |

**Precautions on battery**

The batteries cannot be directly connected in parallel, because the battery's internal resistance is different and it will form a circulation inside the batteries. So it is better to be connected in series. Moreover,a fuse must be set to protect the battery. It is also necessary to consider local weather conditions in case of excessive rainfall and in order to avoid flooding

# SOLAR LIGHT DESIGN FOR NIGHT TIME – AN OVER VIEW

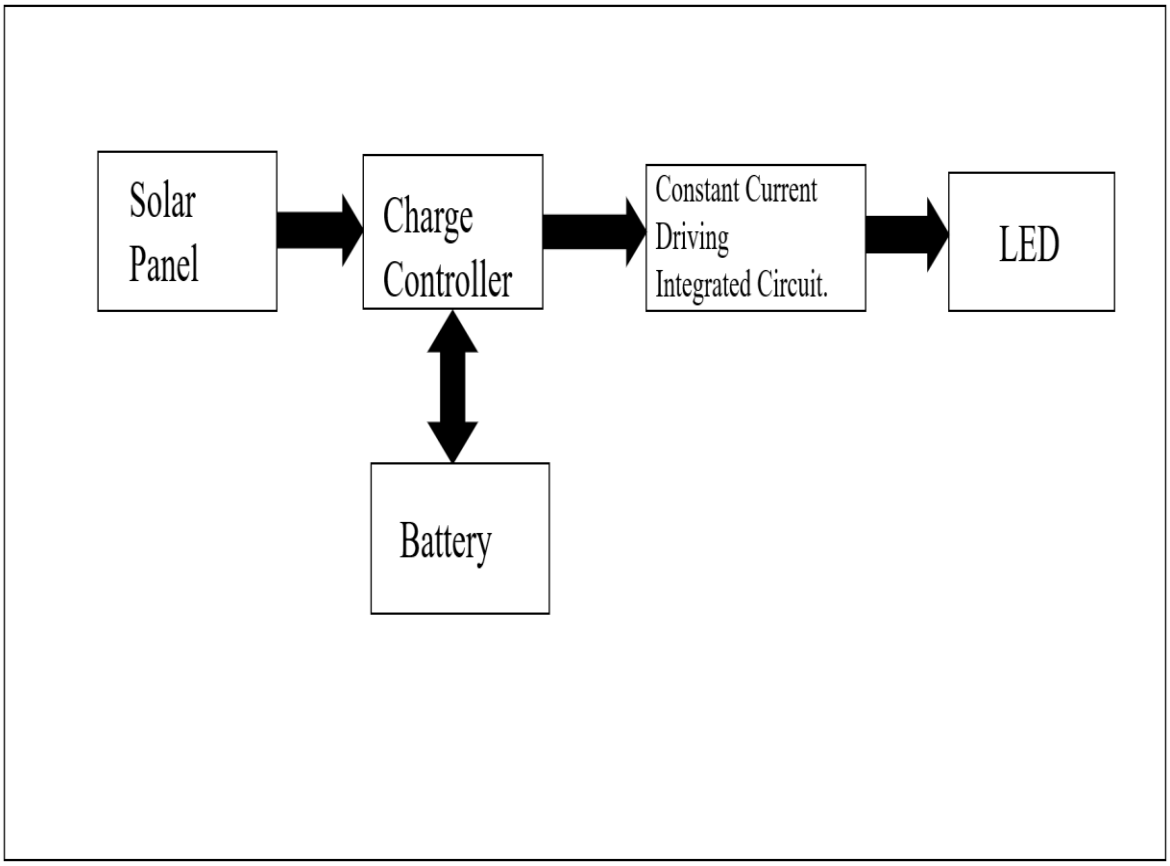
The solar street light does not need to set up the transmission line or route the cable, and no any special management and control are required. It can be installed in the entire public place such as the square, the parking lot, the campus, the street or the highway etc. The street lighting is closely related to people's daily life. Following quick development in process of the global urbanization, the green, efficient, and long-life LED light gradually enters into our lives. Due to photovoltaic effect, solar panel functions to absorb energy through sun throughout the daytime. Here the storage path of the energy to battery passes through the charge-controller. Charge-controller, basically, supervises the flow of current and amount of charging through the solar panel. If this amount exceeds the battery’s capacity, it attempts to protect it from further damage. It has a relay-switching mechanism which switches the path of charging to low current-high resistance path hence the prevention from overcharging is done. Here Dusk-To-Dawn sensor monitors the operation of the circuitry of assembled system. Depending on the light-intensity incident on it, it activates the day-time circuitry i.e. tracking system or night-time circuitry i.e. PIR sensor system. By this way the light operates and helpful for using by glowing when needed with the help of PIR sensor which detects the motion of any person.

The light glows with more voltage when person’s walking under the light whereas glows with less voltage when no one’s passing under it. By this way we can reduce the battery usage upto more extent. The advantages of thus includes low maintenance, power consumption, easy to handle, reduces man work.

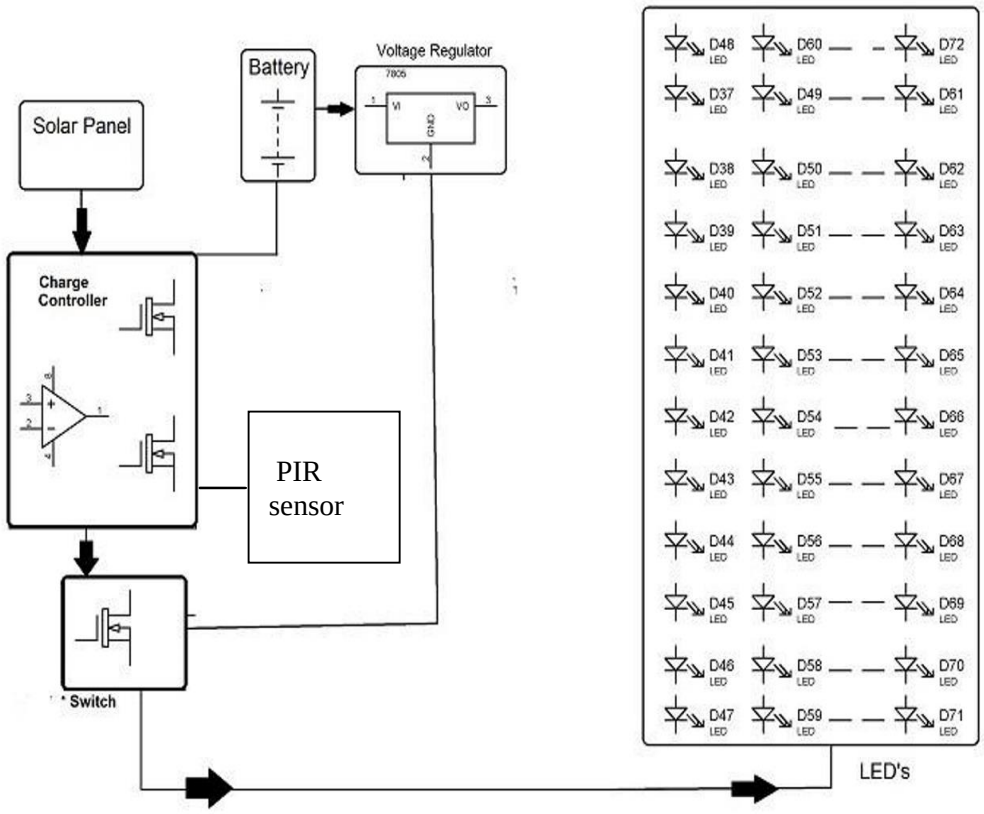
# CHAPTER-4

## DESIGN AND IMPLEMENTATION

## BLOCK DIAGRAM



# CIRCUIT DIAGRAM




Solar charge controllers regulate the energy flowing from the PV array and transfer it directly to the batteries as a DC coupled system, which is the most efficient and effective manner. Controller is a very significant device in solar streetlight, used to decide the status of the charging and lighting by switch on or switch off. A voltage regulator is an electrical regulator designed to automatically maintain a constant voltage level. A voltage regulator may be a simple "feed-forward" design or may include feedback control. It may use an electromechanical mechanism, or electronic components. Depending on the design, it may be used to regulate one or more AC or

DC voltages. A passive infrared sensor is an electronic sensor that measures infrared light radiating from objects in its field of view. They are most often used in PIR-based motion detectors. Whenever there's a person walking near it sense and via charge controller and voltage regulator it makes LED to glow with heavy voltage and without person walking the led glows with less voltage. A rechargeable battery, storage, secondary battery or accumulator is a type of electrical battery which can be charged, discharged into a load, and recharged many times, while a non-rechargeable or primary battery is supplied fully charged, and discarded once discharged. During charging, the battery functions as an electrolytic cell. When the cell is connected to an outside energy source, the electrons in the cathode are forced to flow back to the anode.

## HARDWARE IMPLEMENTATION

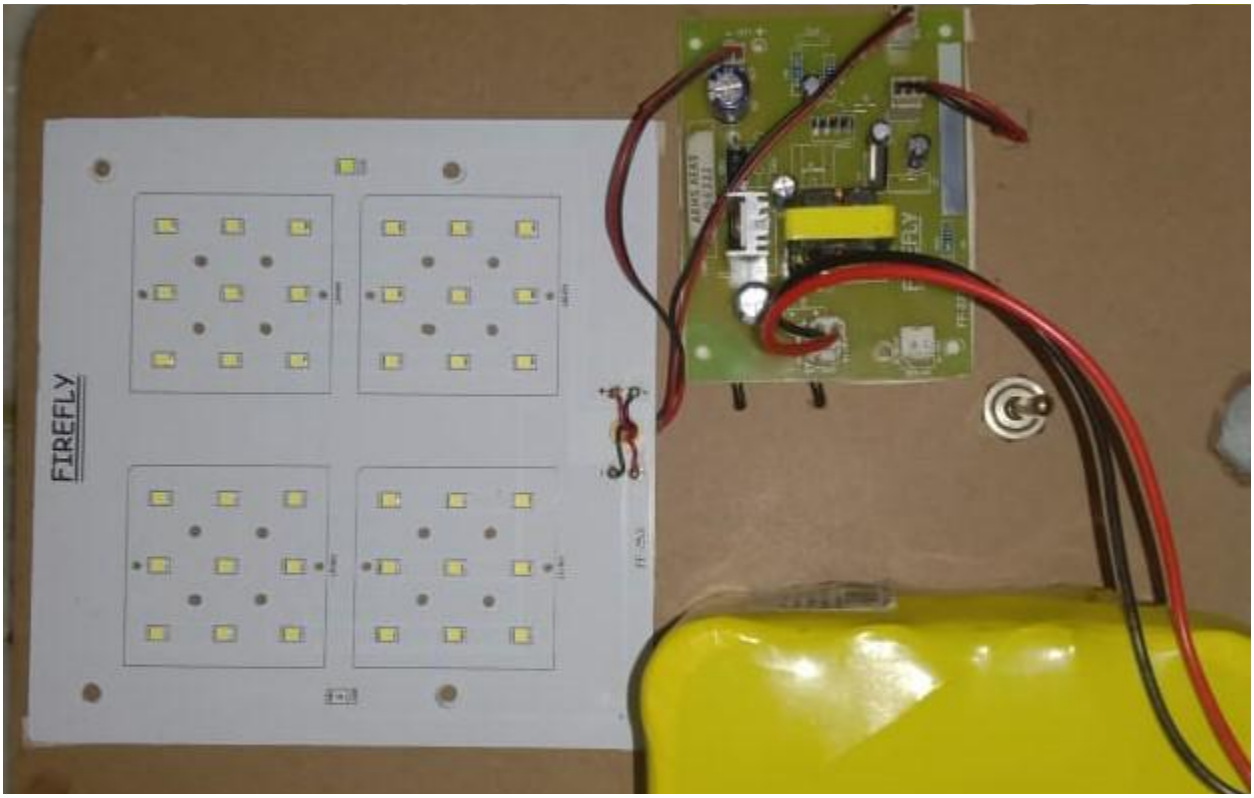
- 12V is the most common solar panel wiring connection with battery. Generally, to achieve the DC to AC both panel and battery are connected in parallel.
- We know that voltage in parallel connection is the same while current is different that is currents are additive in parallel connection.
- In other words the voltage levels of both solar panel and battery remains the same while current capacity would add up.
- We are connecting the negative of solar terminal to the negative of the charge controller likely positive of solar connected to positive of charge controller
- A Voltage Regulator is connected to the controller. It basically controls the voltage of LED without getting any fluctuations in glowing by sending constant voltage.
- Then a PIR sensor is connected to the kit to sense the motion.
- This whole circuitry is then connected to the LED (Light Emitting Diode) bank.

- With the help of mosfet switch we on it and when there is person near the kit the voltage of LED will be high. When there’s no person near the kit over a range of 5 to 10m, the LED will glow with less voltage.

# CHAPTER-5

## RESULTS

The result comprises the successful operation of the ‘SOLAR LIGHT DESIGN FOR NIGHT TIME’. The circuit is stationed in a suitable location that is exposed to sunlight so that immediately it is dark the system automatically switches “ON” the lamps and OFF as we set at daytime as we set time in it. The solar energy is one of the important and major renewable sources of energy and has proven it useful in functioning of applications like streetlights. The charge control is necessary in order to achieve safety and increase the capacity of the battery. In cities, currently thousands of streetlights are operated and the yearly electricity maintenance cost is very high. The initial cost and maintenance can be the drawbacks of this project. With the advancements in technology and good resource, planning the cost of the project can be cut down. Thus, this project can be resourceful and a smart alternative for widely used halogen lamp streetlights in our country.



# CHAPTER-6

## CONCLUSION

Solar energy as one of the important and major renewable source of energy also proves itself useful in functioning of constructions like street-lights. In order to achieve safety and to increase the capacity of battery, charge controller is necessary module. To harness as maximum as possible energy from sun, solar tracker becomes important component. PIR sensor proves itself a quintessential division of given module in order to achieve the aim of minimum consumption of energy as possible. This project gives us an idea of a solar light at night time. This project can be used anywhere either at homes, offices, roads etc. This is also cost efficient.

# CHAPTER-7

## FUTURE SCOPE

Solar energy as one of the important and major renewable source of energy also proves itself useful in functioning of constructions like street-lights. In order to achieve safety and to increase the capacity of battery, charge controller is necessary module. To harness as maximum as possible energy from sun, solar tracker becomes important component. PIR sensor proves itself a quintessential division of given module in order to achieve the aim of minimum consumption of energy as possible. This project gives us an idea of a solar light at night time. This project can be used anywhere either at homes, offices, roads etc. This is also cost efficient.

# CHAPTER-8

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